



Advanced Time Series Analysis and Forecasting

Chapter 4: Cointegration revisited: VECM, ARDL and panel data



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARDL**: AutoRegressive Distributed Lag (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CADF**: Cross-sectionally augmented Dickey–Fuller (regression)
- ▣ **CCE**: Common Correlated Effects (estimator)
- ▣ **CCEMG**: Common Correlated Effects Mean Group
- ▣ **CD**: Cross-section Dependence (test of Pesaran)
- ▣ **CIPS**: Cross-sectionally augmented IPS (test)
- ▣ **CS-ARDL**: Cross-Sectionally augmented ARDL
- ▣ **DOLS**: Dynamic Ordinary Least Squares
- ▣ **ECM**: Error Correction Model
- ▣ **EU**: European Union
- ▣ **EU-27**: the 27 member states of the European Union
- ▣ **FE**: Fixed Effects
- ▣ **FEVD**: Forecast Error Variance Decomposition
- ▣ **FMOLS**: Fully Modified Ordinary Least Squares
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ **GDP:** Gross Domestic Product
- ▣ **GMM:** Generalized Method of Moments
- ▣ **HICP:** Harmonised Index of Consumer Prices
- ▣ **IMF:** International Monetary Fund
- ▣ **IPS:** Im–Pesaran–Shin (panel unit-root test)
- ▣ **LLC:** Levin–Lin–Chu (panel unit-root test)
- ▣ **LLM:** Large Language Model
- ▣ **LR:** Likelihood Ratio (test)
- ▣ **MG:** Mean Group (estimator)
- ▣ **ML:** Machine Learning
- ▣ **NARDL:** Nonlinear AutoRegressive Distributed Lag (model)
- ▣ **NPISH:** Non-Profit Institutions Serving Households
- ▣ **OECD:** Organisation for Economic Co-operation and Development
- ▣ **OLS:** Ordinary Least Squares
- ▣ **PMG:** Pooled Mean Group (estimator)
- ▣ **PPP:** Purchasing Power Parity
- ▣ **PSS:** Pesaran–Shin–Smith
- ▣ **ROBOR:** Romanian Interbank Offer Rate
- ▣ **SE:** Standard Error
- ▣ **TSA:** Time Series Analysis
- ▣ **US:** United States
- ▣ **VAR:** Vector AutoRegression
- ▣ **VECM:** Vector Error Correction Model

Today's question and route

- **Question:** when do non-stationary series share a long-run equilibrium, which restrictions on it can we test, and how far can we trust the answer in samples of 100–300 observations or across 27 countries?
 - ▶ cointegration is a statement about the likelihood of a reduced-rank VAR; inference on it is non-standard and fragile
- **Route** of the chapter
 - ▶ the cointegrated VAR as a likelihood problem; the five deterministic cases; small-sample corrections and the bootstrap
 - ▶ identification and tests on β and α ; interest-rate pass-through in Romania; $I(2)$ in brief; common trends
 - ▶ ARDL and the bounds test; panel unit roots, panel cointegration and heterogeneous panel estimators
- We build on [TSA, Chapter 7](#) (Engle–Granger, ECM, the VECM, Johansen tests, pairs trading) and on [Chapter 3](#) (structural identification); Seminar 4 comes before this lecture

Learning outcomes

- ▣ Derive the Johansen estimator as a reduced-rank regression, state the asymptotic distribution of the trace statistic and choose the deterministic case
- ▣ Correct the rank test in small samples (Reinsel–Ahn, Bartlett, wild bootstrap) and report the sensitivity of the rank to the lag length
- ▣ Identify cointegrating vectors, test economic restrictions on β and weak exogeneity through α , and build a common-trends structural VECM
- ▣ Run the ARDL bounds test with correct (and small-sample) critical values, and decide between ARDL and VECM
- ▣ Test for cross-section dependence, panel unit roots and panel cointegration, and estimate long-run panel relations by MG, PMG and CCE

Reading, data and tools

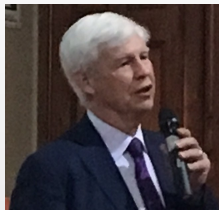
- Backbone: [Johansen \(1995a\)](#); [Juselius \(2006\)](#); [Kilian and Lütkepohl \(2017\)](#), Ch. 3 and 10; [Pesaran \(2015b\)](#), Ch. 22–31; [Lütkepohl \(2005\)](#), Ch. 6–9
 - ▶ original papers: [Johansen \(1988\)](#), [Johansen \(1991\)](#), [Pesaran, Shin and Smith \(2001\)](#), [Pesaran, Shin and Smith \(1999\)](#), [Pesaran \(2006\)](#), [Pesaran \(2007\)](#); survey: [Breitung and Pesaran \(2008\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ Johansen in five cases, restricted ML, bootstrap rank tests, ARDL bounds, LLC/IPS/CIPS, Pedroni, Westerlund, MG, PMG and CCE written out in `numpy`
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
Romanian lending and deposit rates (lei); ROBOR 3M	IMF International Financial Statistics (BNR data); Eurostat irt_st_m	VECM, ARDL
US consumption of nondurables and services, fixed investment, GDP, deflator, population	FRED (PCND, PCESV, FPI, GDP, GDPDEF, B230RC0Q173SBEA)	common trends
Romanian HICP, monthly	Eurostat prc_hicp_minr	I(2)
EU-27 household consumption, disposable income, consumption deflator, population	Eurostat nama_10_gdp, nasa_10_nf_tr, nama_10_pe	panel

- All sources are public and need no account or key; monthly rates to March 2026, US data to 2025, EU annual data to 2024
- HICP: harmonised index of consumer prices; NPISH: non-profit institutions serving households (included with households)

From spurious regressions to a likelihood theory



Robert Engle, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

- 1987: [Engle and Granger \(1987\)](#) define cointegration and prove the representation theorem; estimation is a two-step regression
- 1988–1991: [Johansen \(1988\)](#) and [Johansen \(1991\)](#) turn it into maximum likelihood for a VAR with reduced rank: several relations, tests on their coefficients
- 1999–2001: Pesaran, Shin and Smith add ARDL bounds tests [Pesaran, Shin and Smith \(2001\)](#) and pooled panels [Pesaran, Shin and Smith \(1999\)](#)
- 2003: Engle and Granger share the Nobel Prize; Granger “for methods of analyzing economic time series with common trends (cointegration)”
- Since then: bootstrap rank tests, cross-section dependence in panels, common factors

From TSA to this chapter

- ▣ Known (TSA, Chapter 7): $y_t \sim I(1)$, $\beta' y_t \sim I(0)$; Engle–Granger two-step; the ECM; $\Pi = \alpha\beta'$; trace and max-eigenvalue tests with tabulated critical values
- ▣ New here
 - ▶ why the eigenvalue problem is the maximum likelihood estimator, and what its limit distribution looks like
 - ▶ which deterministic case to use, and what goes wrong in samples of 100–300 months
 - ▶ restrictions on β (economic hypotheses) and on α (weak exogeneity); structural shocks with permanent effects
 - ▶ single equations (ARDL) and panels of countries when the system approach is too demanding

The VECM and the hypothesis $H(r)$ (1/2)

- The VECM: this month's change of each variable responds to last month's levels, to past changes and to deterministic terms

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t$$

- Notation

- ▶ $y_t \in \mathbb{R}^n$: the vector of the n $I(1)$ variables; $\Delta y_t = y_t - y_{t-1}$: their changes
- ▶ Π ($n \times n$): the long-run (levels) matrix; Γ_i ($n \times n$): the short-run matrices; p : the lag order of the VAR in levels
- ▶ D_t : deterministic terms (constant, trend, dummies) with coefficients Φ
- ▶ $\varepsilon_t \sim \text{i.i.d. } N(0, \Omega)$: the innovations, with covariance matrix Ω

- Gaussian likelihood conditional on the first p observations: the estimator is ML, the tests are likelihood ratios Johansen (1988), Johansen (1991)

The VECM and the hypothesis $H(r)$ (2/2)

- $H(r)$: Π has rank at most r , so it factors as $\Pi = \alpha\beta'$ with α, β of dimension $n \times r$
 - ▶ r : the number of cointegrating relations, $0 \leq r \leq n$
 - ▶ β : its columns are the r cointegrating vectors; $\beta'y_t$ is stationary (the long run)
 - ▶ α : the adjustment (loading) coefficients; α_{ij} is the response of $\Delta y_{i,t}$ to the deviation $\beta'_j y_{t-1}$
- The models are nested: $H(0) \subset H(1) \subset \dots \subset H(n)$
 - ▶ $H(n)$: unrestricted VAR in levels; $H(0)$: VAR in differences
- Only Π is identified, not α and β separately: $\alpha\beta' = (\alpha\xi)(\beta\xi^{-1'})'$
 - ▶ ξ : any invertible $r \times r$ matrix; $\xi^{-1'}$: the transpose of its inverse

Step 1: concentrating out the short run

- Regress Δy_t and y_{t-1} on $Z_{2t} = (\Delta y'_{t-1}, \dots, \Delta y'_{t-p+1}, D'_t)'$ and keep the residuals (Frisch–Waugh)
 - ▶ Z_{2t} : all short-run regressors (lagged changes and deterministic terms)
 - ▶ R_{0t}, R_{1t} : the residuals of Δy_t and of y_{t-1} , i.e. both purged of the short run
- The concentrated model is a regression with a reduced-rank coefficient: $R_{0t} = \alpha \beta' R_{1t} + \hat{\varepsilon}_t$
- Moment matrices $S_{ij} = T^{-1} \sum_t R_{it} R'_{jt}$, $i, j \in \{0, 1\}$
 - ▶ T : the number of observations used; S_{00}, S_{11} : the covariance matrices of R_{0t}, R_{1t} ; $S_{01} = S'_{10}$: their cross-covariance
- For fixed β , OLS of R_{0t} on $\beta' R_{1t}$ gives α and Ω in closed form:
 - ▶ $\hat{\alpha}(\beta) = S_{01} \beta (\beta' S_{11} \beta)^{-1}$, $\hat{\Omega}(\beta) = S_{00} - S_{01} \beta (\beta' S_{11} \beta)^{-1} \beta' S_{10}$
 - ▶ $L_{\max}(\beta)$: the likelihood maximised over all parameters except β ; $L_{\max}^{-2/T}(\beta) \propto |\hat{\Omega}(\beta)|$ ($|\cdot|$: determinant)
 - ▶ what remains is a problem in β alone: minimise $|\hat{\Omega}(\beta)|$

Step 2: the eigenvalue problem

- Determinant identity: $|\hat{\Omega}(\beta)| = |S_{00}| \frac{|\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|}{|\beta'S_{11}\beta|}$
- Minimising this ratio of quadratic forms is the generalised eigenvalue problem $|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0$
 - its n roots, ordered, are the eigenvalues $1 > \hat{\lambda}_1 > \dots > \hat{\lambda}_n > 0$; $\hat{v}_i$: the eigenvector of $\hat{\lambda}_i$
- $\hat{\beta}$ = the eigenvectors $\hat{v}_1, \dots, \hat{v}_r$ of the r largest eigenvalues; then $L_{\max}^{-2/T} = |S_{00}| \prod_{i=1}^r (1 - \hat{\lambda}_i)$
 - normalisation $\hat{\beta}' S_{11} \hat{\beta} = I_r$ (I_r : the $r \times r$ identity matrix)
- $\hat{\lambda}_i$ are the squared canonical correlations between R_{0t} and R_{1t} : large $\hat{\lambda}_i$ = a linear combination of levels that predicts the changes
- One eigen-decomposition gives the ML estimates for every r at once (proof: [► Appendix](#))

Likelihood ratio tests of the rank (1/2)

- **Trace:** $H(r)$ against $H(n)$: $LR_{tr}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$
 - ▶ sums the $n - r$ smallest eigenvalues, those that $H(r)$ sets to zero
 - ▶ $\hat{\lambda}_i \approx 0$ gives $-\ln(1 - \hat{\lambda}_i) \approx 0$: a small statistic supports $H(r)$; large values reject it
- **Maximum eigenvalue:** $H(r)$ against $H(r + 1)$: $LR_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
 - ▶ uses only the next eigenvalue, $\hat{\lambda}_{r+1}$: is there one more relation?
- **Rank determination:** test $r = 0, 1, \dots$ in turn and stop at the first non-rejection; this sequence is consistent for the true rank at a fixed level [Johansen \(1995a\)](#)

Likelihood ratio tests of the rank (2/2)

- Limit under $H(r)$:

$$LR_{tr} \Rightarrow \text{tr} \left\{ \int_0^1 (dB) F' \left(\int_0^1 FF' du \right)^{-1} \int_0^1 F (dB)' \right\}$$

- Notation

- ▶ $\Rightarrow$: convergence in distribution as $T \rightarrow \infty$; tr : the trace (sum of the diagonal elements)
- ▶ $B(u)$, $u \in [0, 1]$: an $(n - r)$ -dimensional standard Brownian motion, the limit of the rescaled common trends
- ▶ $F = B$ corrected for the deterministic terms of the case

- The limit depends only on $n - r$ and the case, not on $\alpha, \beta, \Gamma_i, \Omega$

- ▶ a multivariate Dickey–Fuller distribution: quantiles by simulation [Osterwald-Lenum \(1992\)](#), [MacKinnon, Haug and Michelis \(1999\)](#)
- ▶ non-standard: neither χ^2 nor Normal

The Granger representation theorem (1/3)



Clive Granger, 2008

Photo: Olaf Storbeck (2008); CC BY-SA 2.0;

Wikimedia Commons

- The theorem writes a cointegrated VAR as random-walk trends plus stationary deviations
- VAR in levels: $y_t = \sum_{i=1}^p A_i y_{t-i} + \Phi D_t + \varepsilon_t$
 - ▶ A_i ($n \times n$): its lag matrices; $A(z) = I - \sum_{i=1}^p A_i z^i$: its characteristic polynomial
 - ▶ $\Pi = -A(1)$, $\Gamma_i = -\sum_{j>i} A_j$: the VECM matrices; $\Gamma = I - \sum_i \Gamma_i$
- Conditions
 - ▶ the roots of $|A(z)| = 0$ satisfy $|z| > 1$ or $z = 1$ (no explosive or seasonal roots)
 - ▶ $\text{rank } \Pi = r$, so $\Pi = \alpha\beta'$
 - ▶ $\alpha'_\perp \Gamma \beta_\perp$ has full rank $n - r$
 - ▶ $\alpha_\perp$, $\beta_\perp$ ($n \times (n - r)$): orthogonal complements, $\alpha'_\perp \alpha_\perp = 0$, $\beta'_\perp \beta_\perp = 0$

The Granger representation theorem (2/3)

- Under these conditions y_t is $I(1)$, $\beta'y_t$ is $I(0)$, and

$$y_t = C \sum_{i=1}^t (\varepsilon_i + \Phi D_i) + C^*(L)(\varepsilon_t + \Phi D_t) + A_0$$

- Notation

- ▶ $C = \beta_{\perp}(\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp}$: the long-run impact matrix, of rank $n - r$ (derivation: [Appendix](#))
- ▶ $C^*(L) = \sum_{j \geq 0} C_j^* L^j$: a polynomial in the lag operator L ($Ly_t = y_{t-1}$) with summable coefficients, the stationary part
- ▶ A_0 : a constant that depends on the initial values, with $\beta' A_0 = 0$

- First term: random walks plus deterministic trends; second term: stationary deviations from them

The Granger representation theorem (3/3)

- Interpretation
 - ▶ $\beta' C = 0$: the relations do not contain the stochastic trends
 - ▶ $C\alpha = 0$: an equilibrium error has no permanent effect
 - ▶ common trends: the $n - r$ random walks $\alpha'_{\perp} \sum_i \varepsilon_i$
- If $\alpha'_{\perp} \Gamma \beta_{\perp}$ is singular, the theorem fails
 - ▶ C does not exist and some trends are $I(2)$ (see the $I(2)$ section)

Recap: The cointegrated VAR

- Concentrate out the short run, then maximise over β : a generalised eigenvalue problem
- The eigenvalues are squared canonical correlations; the trace and max-eigenvalue statistics are likelihood ratios
- Their limits are Brownian functionals that depend on $n - r$ and on the deterministic terms only
- Granger representation: $C = \beta_{\perp}(\alpha'_{\perp}\Gamma\beta_{\perp})^{-1}\alpha'_{\perp}$ links α, β to the common trends

Five ways to place a constant and a trend

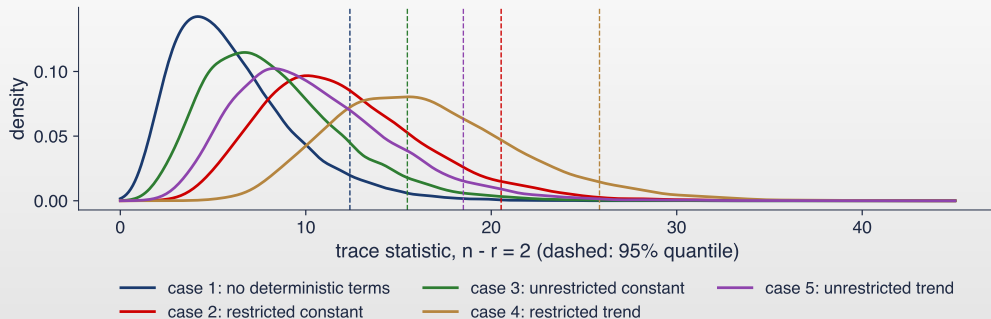
- $\mu_t = \mu_0 + \mu_1 t$ with $\mu_j = \alpha \rho_j + \alpha_{\perp} \gamma_j$: the α part sits in the relations, the $\alpha_{\perp}$ part drives the levels [Johansen \(1995a\)](#)

► μ_t : constant μ_0 plus trend $\mu_1 t$ in the VECM; ρ_j ($r \times 1$), γ_j ($((n-r) \times 1)$): their coordinates in the two directions

	Case	Restriction	Levels y_t	Relations $\beta' y_t$
	1 (H_2)	$\mu_0 = \mu_1 = 0$	no drift	zero mean
□	2 (H_1^*)	$\mu_0 = \alpha \rho_0, \mu_1 = 0$	no drift	constant
	3 (H_1)	$\mu_1 = 0$	linear trend	constant
	4 (H^*)	$\mu_1 = \alpha \rho_1$	linear trend	linear trend
	5 (H)	none	quadratic trend	linear trend

- “Restricted” terms enter Z_{1t} (the regressors multiplied by Π) with y_{t-1} , as part of β ; “unrestricted” terms enter Z_{2t} and are concentrated out
- Interest rates: case 2; trending macro aggregates: case 3 or 4; case 5 is rarely plausible

The case changes the null distribution



- Trace statistic under $H(r)$ with $n - r = 2$, simulated from random walks (10 000 replications, $T = 400$; drift in cases 3 and 5)
- 95% quantiles: 12.38 (case 1), 20.53 (2), 15.48 (3), 25.84 (4), 18.50 (5)

Interpreting the five distributions

- Each restricted deterministic term adds a regressor to the reduced-rank problem and shifts the distribution to the right
- A trend in the levels (cases 3 and 5) makes one direction of the common trend deterministic: for $n - r = 1$ the limit is $\chi^2(1)$, 95% quantile 3.89
- Using case-3 critical values in a case-2 model over-rejects: the 95% quantile for $n - r = 1$ is 9.25 in case 2 and 3.89 in case 3
- Our simulated quantiles differ from the published tables [MacKinnon, Haug and Michelis \(1999\)](#) by a few tenths at most; the Quantlet gives all cases and $n - r = 1, \dots, 4$

Simulated 95% quantiles of the trace statistic

$n - r$	case 1	case 2	case 3	case 4	case 5
1	4.21	9.25	3.89	12.58	3.84
2	12.38	20.53	15.48	25.84	18.50
3	24.44	35.24	29.78	43.14	35.36
4	40.29	53.68	47.98	63.85	55.54

- Simulated with 10 000 replications of a VAR(1) test regression on $T = 400$ observations; the response-surface critical values of [MacKinnon, Haug and Michelis \(1999\)](#) differ by at most a few tenths
- Choosing the case: plot the data first; if unsure between 2 and 3 (or 3 and 4), test the joint hypothesis on rank and case in the order of the Pantula principle [Pantula \(1989\)](#), [Johansen \(1992\)](#)
- Breaks in the deterministic terms shift the distribution again: use the critical values of [Johansen, Mosconi and Nielsen \(2000\)](#) for broken trends

Recap: Deterministic terms

- Split each deterministic term into an α part (in the relations) and an $\alpha_{\perp}$ part (in the levels)
- The five cases have five different null distributions; mixing them up changes the rank
- Plot the data, choose the case on economic grounds and use the Pantula principle when in doubt

Over-rejection of the asymptotic test (1/2)

- The VECM spends many parameters on the short run
 - ▶ $n^2(p-1)$ elements of $\Gamma_1, \dots, \Gamma_{p-1}$, plus the deterministic terms
 - ▶ with $T = 100$, $n = 3$ and $p = 4$: 27 short-run parameters, a large share of the information
- The distortion grows with persistence
 - ▶ persistent short-run dynamics: roots of the stationary part close to the unit circle
 - ▶ the asymptotic test then finds too many cointegrating relations
- Conditional heteroskedasticity [Cavaliere, Rahbek and Taylor \(2010\)](#)
 - ▶ leaves the limit distribution unchanged
 - ▶ but worsens the finite-sample size: use the wild bootstrap

Over-rejection of the asymptotic test (2/2)

▣ **Reinsel–Ahn** Reinsel and Ahn (1992)

- ▶ multiply the statistic by $(T - np)/T$: a degrees-of-freedom correction
- ▶ simple, often too conservative

▣ **Bartlett correction** Johansen (2002)

- ▶ $\mathbb{E}LR_{tr} \approx f(1 + a(\theta)/T)$; the corrected statistic is $LR_{tr}/(1 + a(\theta)/T)$
- ▶ f : the mean of the limit distribution; θ : the VECM parameters
- ▶ $a(\theta)$: an analytic function of θ , evaluated at the estimates

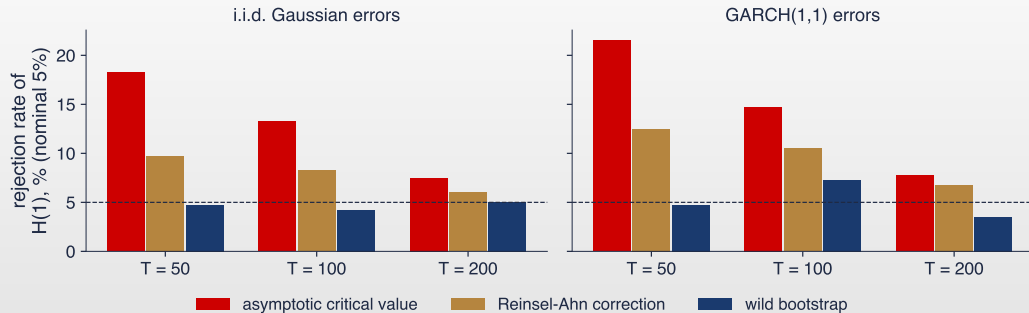
▣ **Bootstrap** Swensen (2006), Cavaliere, Rahbek and Taylor (2012)

- ▶ simulate the null distribution from the model estimated under $H(r)$

The bootstrap rank test, step by step

- ▣ 1. Estimate the VECM under $H(r)$: $\hat{\alpha}^{(r)}, \hat{\beta}^{(r)}, \hat{\Gamma}_i^{(r)}, \hat{\Phi}^{(r)}$ and residuals $\hat{\varepsilon}_t^{(r)}$
- ▣ 2. Generate $\Delta y_t^* = \hat{\alpha}^{(r)} \hat{\beta}^{(r)'} y_{t-1}^* + \sum_i \hat{\Gamma}_i^{(r)} \Delta y_{t-i}^* + \hat{\Phi}^{(r)} D_t + \varepsilon_t^*$ from the observed initial values
- ▣ 3. Innovations: $\varepsilon_t^* = \hat{\varepsilon}_t^{(r)} w_t$, $w_t \sim \text{i.i.d. } N(0, 1)$ (wild) or resampled residuals (i.i.d. bootstrap)
 - the star marks bootstrap quantities; w_t : a random sign-and-scale weight; the wild version keeps the volatility pattern of each date
- ▣ 4. Compute $LR_{tr}^*(r)$ on each sample; p-value = share of $LR_{tr}^* \geq LR_{tr}(r)$; test $r = 0, 1, \dots$ and stop at the first non-rejection
- ▣ Estimating under $H(r)$ (not under $H(n)$) is essential: the bootstrap data must have exactly $n - r$ unit roots
Cavaliere, Rahbek and Taylor (2012)

Size of the rank test in small samples (simulation)



- Three-variable VECM, true rank 1, $\Gamma_1 = 0.8I$, case 2, VAR(2); test of $H(1)$ at 5%; 400 replications, 199 bootstrap samples each
- $T = 50$, i.i.d. errors: asymptotic 18.2%, Reinsel-Ahn 9.8%, wild bootstrap 4.8%

Interpreting the size simulation

- ▣ With persistent short-run dynamics the asymptotic test finds a spurious second relation in 18.2% of samples of 50 and 13.2% of samples of 100
- ▣ Reinsel–Ahn halves the distortion; the wild bootstrap is close to 5% at every T , also under GARCH errors (4.8% at $T = 50$, asymptotic 21.5%)
- ▣ At $T = 200$ all three are acceptable: the problem is the ratio of parameters to observations, not the method
- ▣ Practice: report asymptotic and bootstrap p -values side by side, and the rank for neighbouring lag lengths

Recap: Small samples

- ▣ The asymptotic trace test over-rejects when parameters are many and the short run is persistent
- ▣ Reinsel–Ahn and the Bartlett correction rescale the statistic; the bootstrap rebuilds its distribution
- ▣ Bootstrap from the model estimated under $H(r)$; use wild weights when volatility changes over time

Normalisation is not identification (1/2)

- The eigenvectors give one basis of the cointegration space $\text{sp}(\beta)$; any $\beta\xi$ spans the same space and fits equally well
 - ▶ $\text{sp}(\beta)$: the space spanned by the columns of β ; ξ : an invertible $r \times r$ matrix (a rotation)
- Dividing each vector by one coefficient fixes the scale (r restrictions), but not the rotation
 - ▶ $r(r - 1)$ further restrictions are needed, $r - 1$ per vector
- Identification rests on economic theory: each restriction is a statement such as “the lending rate does not depend on the deposit rate in the long run”

Normalisation is not identification (2/2)

- Linear restrictions vector by vector: $\beta = (H_1\varphi_1, \dots, H_r\varphi_r)$
 - ▶ n_1 : the length of each vector (n plus the restricted deterministic terms)
 - ▶ H_i ($n_1 \times s_i$, known): the design matrix of vector i ; φ_i ($s_i \times 1$): its s_i free coefficients
 - ▶ $R_i = H_{i\perp}$: the restrictions written as $R_i'\beta_i = 0$
- Rank condition [Johansen \(1995b\)](#): for every i and every set of k other vectors $i_1, \dots, i_k$
 - ▶ $\text{rank}(R_i'(H_{i_1}, \dots, H_{i_k})) \geq k$, $k = 1, \dots, r-1$
 - ▶ meaning: no combination of other vectors satisfies the restrictions of vector i
 - ▶ the cointegration analogue of the rank condition in simultaneous equations; the order condition alone is not enough

Testing restrictions on β (1/2)

- **The same restriction on all vectors**, $\beta = H\varphi$: solve $|\lambda H' S_{11} H - H' S_{10} S_{00}^{-1} S_{01} H| = 0$
 - ▶ H ($n_1 \times s$, known): the restriction; φ ($s \times r$): the free coefficients; e.g. excluding one variable from all relations
 - ▶ $\tilde{\lambda}_i$: the eigenvalues of the restricted problem; $\hat{\lambda}_i$: the unrestricted ones
- Test statistic: $LR = T \sum_{i=1}^r \ln\{(1 - \tilde{\lambda}_i)/(1 - \hat{\lambda}_i)\} \rightarrow \chi^2(r(n_1 - s))$
 - ▶ $LR \geq 0$; it is large when the restriction lowers the canonical correlations, i.e. when the data reject it
 - ▶ degrees of freedom $r(n_1 - s)$: $n_1 - s$ restrictions on each of the r vectors
- Given the rank, $\hat{\beta}$ is super-consistent and mixed Gaussian: LR tests on β are χ^2 , unlike the rank test [Johansen \(1991\)](#)

Testing restrictions on β (2/2)

- ▣ **Different restrictions per vector**, $\beta = (H_1\varphi_1, \dots, H_r\varphi_r)$: no closed form
 - ▶ maximise the concentrated log-likelihood $-\frac{T}{2} \ln |\hat{\Omega}(\beta)|$ by the switching algorithm (one vector at a time) or numerically
 - ▶ over-identifying restrictions: $LR \rightarrow \chi^2(\sum_i (n_1 - r + 1 - s_i))$; each vector has $n_1 - r + 1 - s_i$ restrictions beyond the $r - 1$ needed to identify it
- ▣ **Fully known** $\beta = \beta_0$ (e.g. the great ratios):

$$LR = T \left\{ \ln |\hat{\Omega}(\beta_0)| - \ln |S_{00}| - \sum_{i \leq r} \ln(1 - \hat{\lambda}_i) \right\} \rightarrow \chi^2(r(n_1 - r))$$

- ▶ compares the fit with β_0 imposed ($\hat{\Omega}(\beta_0)$) to the unrestricted fit; β_0 : the hypothesised matrix

Restrictions on α and weak exogeneity

- $\alpha = A\psi$: only m combinations of the equations adjust to the relations
 - ▶ A ($n \times m$, known): the restriction; ψ ($m \times r$): the free adjustment coefficients; $A_{\perp}$: its orthogonal complement
 - ▶ condition the system on $A'_{\perp} R_{0t}$ and solve the eigenvalue problem of the remaining m equations (eigenvalues $\tilde{\lambda}_i$)
 - ▶ $LR = T \sum_{i \leq r} \ln\{(1 - \tilde{\lambda}_i)/(1 - \hat{\lambda}_i)\} \rightarrow \chi^2(r(n - m))$
- **Weak exogeneity** of x_t for β : the row of α for x_t is zero (a zero row of α , r restrictions)
 - ▶ x_t does not adjust to past disequilibria; it is a common trend (a column of $\alpha_{\perp}$)
 - ▶ the conditional model of Δy_t given Δx_t is then fully efficient for β : the basis of single-equation ECM and ARDL
- Long-run structural hypotheses are joint restrictions on β and α
 - ▶ PPP: $(1, -1, -1)$ on (price, foreign price, exchange rate); Fisher: $(1, -1)$ on (interest rate, inflation); complete pass-through: $(1, -1)$ on (retail rate, policy or market rate)

Recap: Identification and tests

- ▣ Identify β with $r - 1$ economically motivated restrictions per vector; check the rank condition
- ▣ Given the rank, LR tests on β and α are χ^2 with known degrees of freedom
- ▣ A zero row of α = weak exogeneity = permission to condition on that variable

How fast and how fully do banks follow ROBOR?



The BNR palace, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia

[Commons](#)

- Monetary policy reaches households and firms only through bank rates; ROBOR 3M is the reference for most variable-rate loans in lei
- Euro area: long-run pass-through to lending rates often complete, to deposit rates incomplete [de Bondt \(2005\)](#); CEE: faster and fuller pass-through after the early transition [Égert, Crespo-Cuaresma and Reininger \(2007\)](#)
- Hypotheses: H_β : lending and deposit rates move one for one with ROBOR in the long run; H_α : ROBOR does not adjust to bank rates
- Data: August 2005 – March 2026, $T = 248$ months (inflation targeting)

Lending rate, deposit rate and ROBOR 3M



- Monthly, % per year; lending and deposit rates in lei (IMF IFS, reported by the BNR); ROBOR 3M (Eurostat)
- ROBOR ranges from 0.51% to 18.2%; the lending–ROBOR spread falls from 11.3 pp to 2.5 pp

Interpreting the three rates

- All three rates wander far from any mean over twenty years: treat them as $I(1)$ within the sample, even if interest rates are bounded in theory
- They move together after 2008, in 2014–2015 and in 2022: the visual case for cointegration
- The falling spread in 2005–2010 is the end of the transition (competition, falling risk premia), not a policy effect
- No trend in levels is plausible for rates: case 2, constant restricted to the relations

Rank of the pass-through system

$H(r)$	LR_{tr}	RA	cv 5%	wild p	$LR_{tr}, p=4$	RA, $p=4$	wild $p, p=4$
$r = 0$	121.1	118.1	35.2	0.007	40.0	38.0	0.647
$r \leq 1$	52.2	50.9	20.5	0.022	14.9	14.1	0.812
$r \leq 2$	4.3	4.2	9.3	0.767	3.8	3.6	0.820

- Case 2; RA: Reinsel–Ahn; cv: simulated 95% quantile; wild p : 399 wild-bootstrap samples under $H(r)$; lag order by BIC = 2, HQ = 5, AIC = 8
- With $p = 2$ all three methods give $r = 2$; with $p = 4$ the asymptotic test gives $r = 1$ and the wild bootstrap rejects nothing

Interpreting the rank tests

- ▣ Two relations are what theory predicts: one for lending rates, one for deposit rates, both anchored to ROBOR
- ▣ The result is fragile: two extra lags and heteroskedasticity-robust inference remove the evidence; the 2008–2009 volatility dominates the sample
- ▣ We continue with $r = 2$ because theory and the parsimonious model agree, and we report the sensitivity
- ▣ A rank chosen this way is a modelling decision, not a measurement: state it before testing restrictions

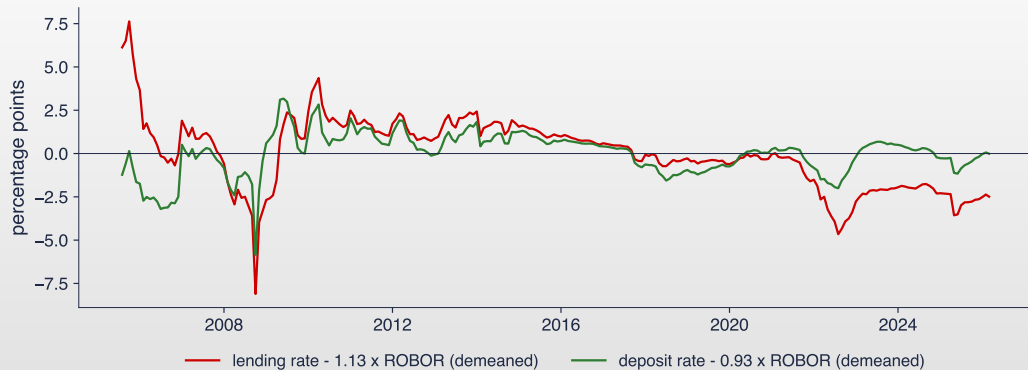
Identified relations and the tests

- Just-identified (one exclusion per vector): $\text{lend}_t = 1.13 \text{ROBOR}_t + 3.62$, $\text{dep}_t = 0.93 \text{ROBOR}_t + 0.09$
- Adjustment: lending rate $\alpha_{11} = -0.056$ (to its own relation), deposit rate $\alpha_{22} = -0.071$; ROBOR: 0.014 and 0.017

	Hypothesis	LR	df	p
□	complete pass-through to both rates	8.98	2	0.011
	complete pass-through to the lending rate	4.16	1	0.041
	ROBOR weakly exogenous (zero row of α)	0.57	2	0.752

- With $p = 4$: lending pass-through 1.11, p -value of complete pass-through to lending 0.458, of weak exogeneity 0.819

The two equilibrium errors



□ $\hat{\beta}'_i y_t$ for the just-identified relations, demeaned; they should look stationary if the rank and β are right

Interpreting the pass-through VECM

- Lending rates over-react slightly in the long run (1.13 per point of ROBOR), deposit rates under-react (0.93): the bank margin widens when ROBOR rises
- Joint complete pass-through is rejected ($p = 0.011$), driven by deposits; for lending rates alone the verdict depends on the lag length
- ROBOR is weakly exogenous ($p = 0.752$): money-market rates lead, bank rates follow; this justifies a single-equation ARDL
- The lending equilibrium error is persistent before 2010: the transition-era spread is not a stable markup, a candidate for a broken constant [Johansen, Mosconi and Nielsen \(2000\)](#)

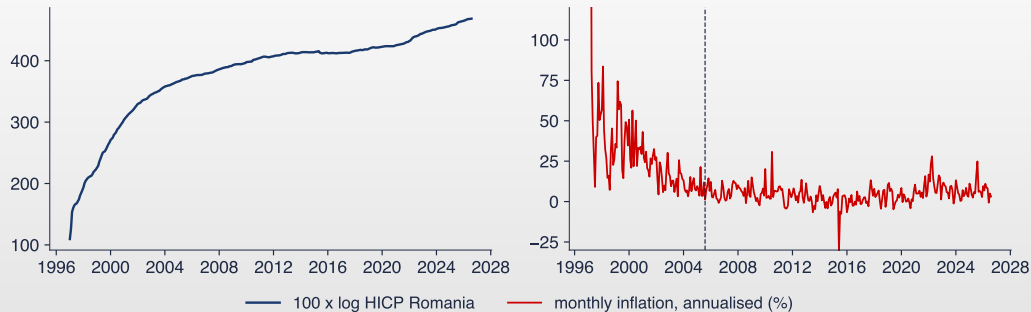
Recap: Pass-through in Romania

- Two cointegrating relations with ROBOR as the common trend, but the rank is sensitive to lags and to heteroskedasticity
- Lending pass-through close to (slightly above) one, deposit pass-through below one
- Weak exogeneity of ROBOR is not rejected: a conditional single-equation model is legitimate

When the I(1) model is not enough

- If $\alpha'_{\perp} \Gamma \beta_{\perp}$ has reduced rank $s < n - r$, the Granger representation fails
 - ▶ some common trends are I(2): y_t contains double sums $\sum_{j \leq t} \sum_{i \leq j} \varepsilon_i$
 - ▶ typical candidates: nominal prices and nominal money when inflation changes; inflation is then I(1)
- The I(2) model [Johansen \(1995c\)](#), [Juselius \(2006\)](#)
 - ▶ two reduced-rank conditions: $\Pi = \alpha \beta'$ and $\alpha'_{\perp} \Gamma \beta_{\perp} = \xi \eta'$
 - ▶ $\xi, \eta ((n - r) \times s, \text{full rank})$: the factors of the second reduced-rank matrix
 - ▶ polynomial cointegration: $\beta' y_t + \delta' \Delta y_t \sim I(0)$; $\delta (n \times r)$: the weights of the differences
 - ▶ example: real money plus a multiple of inflation
- Practical check
 - ▶ test Δy_t for a unit root; a root near one in the stationary part of the I(1) VECM is a warning
 - ▶ the usual fix: model real (deflated) variables and inflation

Is the Romanian price level I(2)?



- Left: 100 ln HICP; right: monthly inflation, annualised; dashed line: inflation targeting from August 2005
- ADF t on monthly inflation (12 lags): -4.25 for 1997–2026, -2.39 since 2005 (5% critical value -2.87); on its change: -6.13

Interpreting the I(2) check

- ▣ Over 1997–2026, disinflation looks like mean reversion and inflation tests as stationary: the price level is I(1)
- ▣ Since 2005 a unit root in inflation is not rejected: within the inflation-targeting sample the price level behaves like I(2)
- ▣ The answer depends on the sample, as for the rank: this is why systems with nominal levels are usually rewritten in real terms and inflation
- ▣ Our pass-through system uses rates, which are at most I(1): the I(2) issue does not arise there

Permanent and transitory shocks (1/2)

- Granger representation: the long-run effect of ε_t on y_{t+h} , $h \rightarrow \infty$, is $C\varepsilon_t$, and $\text{rank } C = n - r$
- Structural shocks: $\varepsilon_t = Bu_t$, $\mathbb{E}u_t u_t' = I$
 - ▶ u_t : n uncorrelated shocks with unit variance; B ($n \times n$): their impact effects, $BB' = \Omega$
 - ▶ the long-run impact CB has at most $n - r$ non-zero columns: $k = n - r$ permanent and r transitory shocks
- Counting (Chapter 3 needed $n(n - 1)/2$ restrictions): cointegration gives rk zero long-run restrictions for free
 - ▶ still needed: $k(k - 1)/2$ among the permanent shocks and $r(r - 1)/2$ among the transitory shocks Kilian and Lütkepohl (2017), Gonzalo and Ng (2001)

Permanent and transitory shocks (2/2)

- With one common trend ($k = 1$) the permanent shock is identified without further assumptions
 - ▶ the shock: $u_t^P = \alpha'_\perp \varepsilon_t / (\alpha'_\perp \Omega \alpha_\perp)^{1/2}$, the innovation of the common trend scaled to unit variance
 - ▶ its impact: $b^P = \Omega \alpha_\perp (\alpha'_\perp \Omega \alpha_\perp)^{-1/2}$, the column of B that moves the n variables on impact
- This generalises Blanchard–Quah ([Chapter 3](#)): there the long-run zero was imposed; here it comes from the cointegration rank

Case study: King, Plosser, Stock and Watson (1991) (1/2)

□ King, Plosser, Stock and Watson (1991)

- ▶ a real business-cycle model with a stochastic productivity trend
- ▶ consumption, investment and output share one common trend
- ▶ the great ratios $c - y$ and $i - y$ are stationary (c, i, y : logarithms per capita)

□ In VECM terms

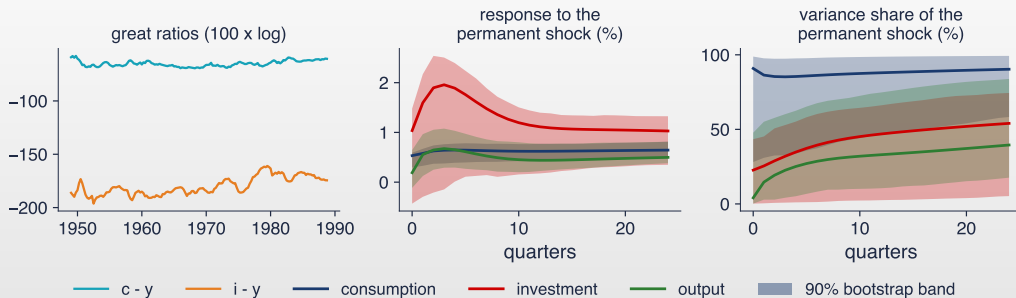
- ▶ $n = 3, r = 2, \beta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$ imposed for (c, i, y)
- ▶ $k = n - r = 1$: one permanent “balanced-growth” shock

□ Question: how much of the business cycle does the permanent shock explain?

Case study: King, Plosser, Stock and Watson (1991) (2/2)

- Our replication: US quarterly data, 1949Q1–1988Q4 (the paper's sample), $T = 160$
 - ▶ c : nondurables and services; i : fixed investment; y : GDP
 - ▶ all per capita, deflated by the GDP deflator
- Rank
 - ▶ VAR(2) in levels (all criteria), case 3
 - ▶ trace: 45.8, 19.4, 1.4 against 95% quantiles 29.8, 15.5, 3.9: $r = 2$
- Test of $\beta = \beta_0$ (the exact great ratios)
 - ▶ case 3: rejected, $LR = 16.6$, $\chi^2(2)$, $p < 0.001$
 - ▶ with a restricted trend (case 4): not rejected, $LR = 0.50$, $p = 0.778$

The balanced-growth shock



- Great ratios imposed (case 3); one-standard-deviation permanent shock; 90% residual-bootstrap bands (300 samples, β fixed)
- Long-run effect 0.74% on all three; share in output variance: 4% at 1 quarter, 30% at 8, 39% at 24 [17; 83]

Interpreting the common-trends model

- Balanced growth holds by construction in the long run: the shock moves c , i and y by the same 0.74%; investment overshoots in the short run
- The permanent shock dominates consumption (87% at 8 quarters, the permanent-income logic) but explains a minority of output fluctuations at business-cycle horizons
- Bands are wide: with $T = 160$ the data barely separate permanent from transitory variation
- As in [Chapter 3](#), the economics is in the identifying assumption: here, which relations are stationary

How robust is the answer?

Specification	$h = 1$	$h = 4$	$h = 8$	$h = 24$
great ratios, case 3, 1949–1988	4%	23%	30%	39%
great ratios with trend, case 4, 1949–1988	34%	37%	52%	77%
great ratios, case 3, 1949–2019	51%	30%	29%	40%
great ratios, case 3, 1949–2025	28%	20%	20%	29%

- Share of the permanent shock in the forecast error variance of output, h quarters ahead
- Allowing the ratios to trend (services gaining share) raises the role of the permanent shock considerably at short horizons; extending the sample changes it again; the great-ratio LR rises to 24.3 on data to 2025
- Conclusion: the size of “trend shocks” in the cycle is not pinned down by the data alone

Recap: Structural VECM

- Cointegration splits shocks into $n - r$ permanent and r transitory ones and supplies rk long-run zeros
- With one common trend the permanent shock is $\alpha'_{\perp} \varepsilon_t$, scaled
- KPSW: balanced growth gives a clean identification, but the variance shares depend on the deterministic case and the sample

From a VECM to a conditional ECM (1/2)

- Partition $y_t = (y_t, x_t')'$, x_t of dimension k ; if x_t is weakly exogenous and there is one relation involving y_t , the conditional model is

$$\Delta y_t = c_0 + c_1 t + \pi_{yy} y_{t-1} + \pi'_{yx} x_{t-1} + \sum_{i=1}^{p-1} \psi_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \omega'_j \Delta x_{t-j} + u_t$$

- Notation Pesaran, Shin and Smith (2001)

- ▶ y_t : the dependent variable; x_t : the k forcing variables; c_0, c_1 : intercept and trend coefficient
- ▶ π_{yy} (scalar), π_{yx} ($k \times 1$): the coefficients on the lagged levels, i.e. the long-run part
- ▶ ψ_i, ω_j : short-run coefficients; p, q : lag orders; u_t : the error, uncorrelated with Δx_t

From a VECM to a conditional ECM (2/2)

- This is an $ARDL(p, q)$ in levels, rewritten
 - ▶ long-run coefficients $\theta = -\pi_{yx}/\pi_{yy}$: the equilibrium is $y = \theta'x$ (plus constant)
 - ▶ speed of adjustment $\pi_{yy} \in (-1, 0)$: the share of last period's disequilibrium corrected this period; half-life $\ln 0.5 / \ln(1 + \pi_{yy})$
- Inference
 - ▶ OLS is consistent and $\hat{\theta}$ has a mixed-Gaussian limit when the lags absorb the endogeneity of x_t Pesaran and Shin (1998)
 - ▶ standard errors of $\hat{\theta}$ by the delta method; t -tests on θ are asymptotically valid
- The attraction: x_t may be $I(0)$, $I(1)$ or a mix, and we need not pre-test it

The bounds test (1/2)

- H_0 : no level relationship, $\pi_{yy} = 0$ and $\pi_{yx} = 0$
 - ▶ F : the Wald statistic of these $k + 1$ restrictions, divided by $k + 1$
 - ▶ t : the t ratio of $\hat{\pi}_{yy}$; it excludes the degenerate case $\pi_{yy} = 0 \neq \pi_{yx}$
- The null distribution depends on the integration order of x_t
 - ▶ lower bound: all of x_t is $I(0)$; upper bound: all of x_t is $I(1)$
- Decision
 - ▶ F above the upper bound: reject, there is a level relationship
 - ▶ F below the lower bound: do not reject
 - ▶ between the bounds: inconclusive, the order of integration of x_t matters

The bounds test (2/2)

- ▣ Five cases, as in Johansen
 - ▶ I: no deterministic terms; II: restricted intercept
 - ▶ III: unrestricted intercept; IV: unrestricted intercept, restricted trend
 - ▶ V: unrestricted intercept and trend
 - ▶ in cases II and IV the restricted terms are part of H_0 ; the t -bounds exist for cases I, III, V
- ▣ Simulated 5% bounds, $k = 1$ ($T = 1000$, 10 000 replications)
 - ▶ F : case II [3.58; 4.14], case III [4.97; 5.71]
 - ▶ t , case III: [-2.86; -3.22]

Critical values in small samples

- ▣ PSS tabulate asymptotic bounds ($T = 1000$); with 30–80 annual observations they are too low [Narayan \(2005\)](#)
- ▣ Our simulation, case III, $k = 1$, 5%: $T = 30$: [5.52; 6.32], $T = 80$: [5.02; 5.86], $T = 250$: [4.98; 5.73], asymptotic: [4.97; 5.71]
- ▣ Response-surface regressions give critical values and approximate p -values for any T , k and lag order [Kripfganz and Schneider \(2020\)](#): use them instead of the asymptotic table
- ▣ Frequent misuses in applied work
 - ▶ the wrong case (case III bounds for a model estimated in case II); I(2) regressors, for which the bounds are invalid
 - ▶ feedback from y_t to x_t (no weak exogeneity), or more than one relation among the variables

ARDL or VECM?

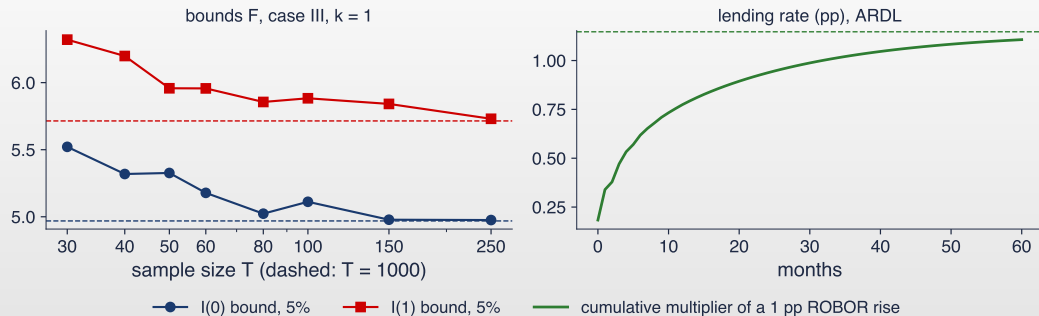
	ARDL	VECM
Relations	one, normalised on y_t	r , rank estimated
Regressors	$I(0)$, $I(1)$ or mixed; weakly exogenous	all $I(1)$; all endogenous
Inference	bounds F and t ; θ by the delta method	trace/max-eigenvalue; χ^2 tests on β , α
Small samples	few parameters; small-sample bounds needed	many parameters; bootstrap needed
Structural use	multipliers of x on y	common trends, identified shocks

- Rule: estimate the VECM first when you can
- ▶ weakly exogenous regressors and a unique relation: the ARDL is efficient and simpler
 - ▶ feedback from y_t to x_t : the ARDL estimates are inconsistent

ARDL pass-through of ROBOR to the lending rate

- ▣ ARDL(4, 2) selected by BIC, case II (ROBOR weakly exogenous in the VECM), $n = 244$
- ▣ Bounds $F = 4.27$; 5% bounds asymptotic $[3.58; 4.14]$, at our T $[3.67; 4.14]$: a level relationship
 - ▶ the same regression in case III: $F = 5.87$ against $[4.94; 5.79]$: inconclusive; $t = -3.28$ against $[-2.86; -3.22]$
- ▣ Long-run pass-through $\hat{\theta} = 1.15$ (SE 0.12): complete pass-through not rejected by the t -test; speed $\hat{\pi}_{yy} = -0.026$ (SE 0.008), half-life 27 months
- ▣ Deposit rate: $F = 7.68$, $\hat{\theta} = 0.94$ (SE 0.05), half-life 12 months: incomplete but faster

Small-sample bounds and the dynamic multipliers



- Left: simulated 5% bounds, case III, $k = 1$, against T (dashed: $T = 1000$); right: cumulative response of the lending rate to a permanent 1 pp rise of ROBOR (dashed: $\hat{\theta}$)
- Multipliers: 0.18 on impact, 0.62 after 6 months, 0.77 after 12, 0.94 after 24, 1.11 after 60

Interpreting the ARDL results

- The upper bound falls from 6.32 at $T = 30$ to 5.73 at $T = 250$: with annual data the asymptotic table over-rejects
- The verdict on a level relationship depends on the case: decide case II or III from economics (no drift in rates) before looking at F
- ARDL and VECM agree on the size of the lending pass-through (1.15 and 1.13) but not on the test of $\theta = 1$: different nuisance parameters, different power
- About half of the long-run effect arrives within six months (0.62); the rest is slow, with a half-life of the disequilibrium of 27 months, consistent with fixed-rate periods and contract repricing

Two related tools (1/2)

▣ **Toda–Yamamoto causality** Toda and Yamamoto (1995)

- ▶ a VAR of possibly integrated or cointegrated variables
- ▶ fit $p + d_{\max}$ lags in levels and test only the first p ; $d_{\max}$: the highest suspected order of integration
- ▶ the Wald statistic is asymptotically $\chi^2(p)$ whatever the integration and cointegration properties
- ▶ the price: lower power than a correctly specified VECM

▣ **Nonlinear ARDL** Shin, Yu and Greenwood-Nimmo (2014)

- ▶ partial sums of increases: $x_t^+ = \sum_{j \leq t} \max(\Delta x_j, 0)$
- ▶ partial sums of decreases: $x_t^- = \sum_{j \leq t} \min(\Delta x_j, 0)$, so that $x_t = x_0 + x_t^+ + x_t^-$

Two related tools (2/2)

- Nonlinear ARDL: x_t^+ and x_t^- as two regressors
 - ▶ x_t^+ and x_t^- enter the ARDL as two regressors, with long-run coefficients θ^+ , θ^- and short-run coefficients ω_j^+ , ω_j^-
 - ▶ long-run asymmetry: $\theta^+ \neq \theta^-$; short-run asymmetry: different ω_j^+ and ω_j^-
 - ▶ both tested by Wald tests; the bounds test uses $k = 2$
- Classic use: “rockets and feathers” [Bacon \(1991\)](#)
 - ▶ retail fuel prices rise fast with oil and fall slowly
 - ▶ Seminar 4 tests it for Romania

Recap: ARDL and the bounds test

- ▣ The conditional ECM is an ARDL rewritten; it needs weak exogeneity and a single relation
- ▣ The bounds test avoids pre-testing the regressors, at the price of an inconclusive zone
- ▣ Use the right case and small-sample (response-surface) critical values
- ▣ Toda–Yamamoto for causality without pre-testing; NARDL for asymmetric adjustment

Panels: the gain and the new problems



The ECB seat, Frankfurt

Photo: DXR (2015); CC BY-SA 4.0;

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- The gain: N countries add information
 - ▶ country series are short: 20–30 years of annual data
 - ▶ a single-country rank or bounds test has little power
- Asymptotics in two dimensions [Phillips and Moon \(1999\)](#), [Kao \(1999\)](#)
 - ▶ sequential ($T \rightarrow \infty$, then $N \rightarrow \infty$) or joint, with $N/T \rightarrow 0$
 - ▶ a panel regression of $I(1)$ series estimates an average long-run relation even without cointegration
- Two new problems
 - ▶ **heterogeneity**: the dynamics differ across countries
 - ▶ **cross-section dependence**: common shocks (2009, COVID-19, the euro)
 - ▶ first-generation tests assume independent units; second-generation tests model the dependence [Breitung and Pesaran \(2008\)](#)

Measuring cross-section dependence

- **CD statistic** [Pesaran \(2021\)](#): the scaled sum of all pairwise correlations

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i < j} \hat{\rho}_{ij} \rightarrow N(0, 1)$$

- ▶ $\hat{\rho}_{ij}$: the correlation over time between the residuals (or demeaned series) of units i and j ; N : number of units; T : number of periods
- ▶ reading: $CD \approx 0$ under the null; $|CD| > 1.96$ rejects it at 5%
- The null is *weak dependence* [Pesaran \(2015a\)](#)
 - ▶ correlation that vanishes on average as N grows
 - ▶ a common factor with non-zero mean loadings gives $|CD| \rightarrow \infty$
 - ▶ valid for fixed T and large N , also in dynamic and unit-root panels
 - ▶ positive and negative correlations can offset each other: report the mean $|\hat{\rho}_{ij}|$ as well

Cross-section dependence in EU-27 data

- EU-27, $T = 22$ (2001–2022)
 - ▶ consumption growth: $CD = 50.3$, mean $|\hat{\rho}_{ij}| = 0.59$
 - ▶ income growth: $CD = 25.7$; inflation: $CD = 62.4$
- Interpretation
 - ▶ every statistic is far above 1.96: strong dependence everywhere
 - ▶ first-generation panel tests are not reliable for these data

Panel unit-root tests (1/2)

- ▣ **LLC** Levin, Lin and Chu (2002): one ADF regression per unit with a common ρ

$$\Delta y_{it} = \mu_i + \rho y_{i,t-1} + \sum_j \gamma_{ij} \Delta y_{i,t-j} + e_{it}$$

- ▶ y_{it} : unit i at time t ; μ_i : unit intercepts; γ_{ij} : unit-specific lag coefficients; e_{it} : errors
- ▶ H_0 : $\rho = 0$ (unit root in every unit); H_1 : $\rho < 0$, all units stationary with the same ρ
- ▶ pooled t on orthogonalised, variance-normalised residuals, adjusted by tabulated mean and variance

- ▣ **IPS** Im, Pesaran and Shin (2003): heterogeneous ρ_i ; average the individual ADF statistics t_i and standardise

$$\bar{t} = N^{-1} \sum_i t_i, \quad Z = \sqrt{N}(\bar{t} - \mathbb{E}t) / \sqrt{\text{Var}(t)} \rightarrow N(0, 1)$$

- ▶ $\mathbb{E}t$, $\text{Var}(t)$: the mean and variance of one ADF t under the null (tabulated); a very negative Z rejects
- ▶ H_1 : a non-zero fraction of units is stationary; rejection does not say which

Panel unit-root tests (2/2)

- ▣ **CIPS Pesaran (2007)**: one common factor f_t in the errors; augment each ADF with $\bar{y}_{t-1}$ and $\Delta\bar{y}_{t-j}$ (CADF), average the t -ratios
 - ▶ $\bar{y}_t = N^{-1} \sum_i y_{it}$: the cross-section average at time t ; f_t : an unobserved shock common to all units
 - ▶ the cross-section averages proxy f_t ; the distribution is non-standard and tabulated
- ▣ Under strong dependence LLC and IPS over-reject: they treat 27 correlated countries as 27 independent pieces of evidence

Panel unit roots in EU-27 data

Variable	LLC	IPS $\bar{t}$	CIPS
consumption (trend)	-12.85 (0.075)	-2.67 (0.003)	-2.38 (0.290)
income (trend)	-11.82 (0.305)	-2.09 (0.660)	-2.18 (0.566)
inflation (constant)	-10.10 (0.003)	-1.82 (0.038)	-2.36 (0.009)

- c, y : 100 ln real per capita household consumption and disposable income; π : consumption-deflator inflation; $N = 27$, $T = 22$, one lag
- p -values in brackets from 1 000 simulations of the null with independent units, for our N and T (replacing the tabulated moments)
- Consumption: IPS rejects the unit root ($p = 0.003$), CIPS does not ($p = 0.290$): the “stationarity” was a common factor; inflation is stationary by every test

Panel cointegration tests (1/2)

- ▣ **Residual-based** Pedroni (1999), Pedroni (2004): estimate $y_{it} = a_i + \delta_i t + \beta_i' x_{it} + e_{it}$ country by country and test the residuals for a unit root
 - ▶ a_i, δ_i : country intercept and trend; β_i : country cointegrating coefficients; e_{it} : the equilibrium error, $I(1)$ under H_0
 - ▶ seven statistics: four “panel” (within, common AR coefficient) and three “group” (between, heterogeneous); H_0 : no cointegration in any unit
 - ▶ they impose a common-factor restriction (short-run dynamics of x and y alike); Kao (1999) is the homogeneous special case
- ▣ **ECM-based** Westerlund (2007): test whether each country's ECM has error correction

$$\Delta y_{it} = d_i + \alpha_i y_{i,t-1} + \lambda_i' x_{i,t-1} + \sum_j a_{ij} \Delta y_{i,t-j} + \sum_j \gamma_{ij}' \Delta x_{i,t-j} + e_{it}$$

- ▶ α_i : the speed of adjustment of country i ; λ_i : its level coefficients on x ; d_i, a_{ij}, γ_{ij} : deterministic and short-run terms
- ▶ H_0 : $\alpha_i = 0$ for all i (no error correction anywhere); negative statistics reject

Panel cointegration tests (2/2)

□ Westerlund statistics

- ▶ group: $G_t = N^{-1} \sum_i \hat{\alpha}_i / \text{SE}(\hat{\alpha}_i)$, $G_a = N^{-1} \sum_i T \hat{\alpha}_i / \hat{\alpha}_i(1)$; H_1 : some countries adjust
- ▶ $\hat{\alpha}_i(1) = 1 - \sum_j \hat{a}_{ij}$: scales $\hat{\alpha}_i$ to the long-run adjustment
- ▶ panel: P_t , P_a , the same ratios for a common α ; H_1 : all countries adjust
- ▶ bootstrap critical values under cross-section dependence

□ EU-27, c on y and π (null distributions simulated)

- ▶ Pedroni group ADF (the mean of the country ADF t -ratios of $\hat{e}_{it}$): -3.05 , 5% critical value -2.73 , $p < 0.001$
- ▶ Westerlund G_t : -1.66 , 5% critical value -2.18 , $p = 0.714$

□ The two families disagree

- ▶ the residual-based test rejects, the ECM test does not
- ▶ with $T = 22$ and strong dependence, neither verdict is secure

Estimators of the long-run relation (1/3)

▣ **Panel FMOLS/DOLS** Pedroni (2001)

- ▶ FMOLS: fully modified OLS, corrected for endogeneity and serial correlation Phillips and Hansen (1990)
- ▶ DOLS: OLS with leads and lags of Δx Saikkonen (1991), Stock and Watson (1993)
- ▶ estimated per country and averaged (group mean)

▣ **Mean group (MG)** Pesaran and Smith (1995)

- ▶ estimate each country's ARDL and average the long-run coefficients: $\hat{\theta}_{MG} = N^{-1} \sum_i \hat{\theta}_i$
- ▶ consistent under full heterogeneity; $SE = s_\theta / \sqrt{N}$, s_θ : the cross-country standard deviation of $\hat{\theta}_i$
- ▶ pooled fixed effects with common slopes are inconsistent when the true slopes differ, even for large T ; for small T the Nickell bias adds Nickell (1981)

Estimators of the long-run relation (2/3)

▣ CCE Pesaran (2006)

- ▶ model: $y_{it} = \alpha_i + \beta_i' x_{it} + \gamma_i' f_t + e_{it}$
- ▶ estimation: add the cross-section averages $\bar{y}_t, \bar{x}_t$ to each regression
- ▶ f_t : unobserved common factors; γ_i : the country loadings on them; the averages $\bar{y}_t, \bar{x}_t$ absorb f_t
- ▶ valid with non-stationary factors [Kapetanios, Pesaran and Yamagata \(2011\)](#); dynamic version CS-ARDL [Chudik and Pesaran \(2015\)](#)

Estimators of the long-run relation (3/3)

- ▣ **Pooled mean group (PMG)** Pesaran, Shin and Smith (1999): a common long run inside country-specific ECMs

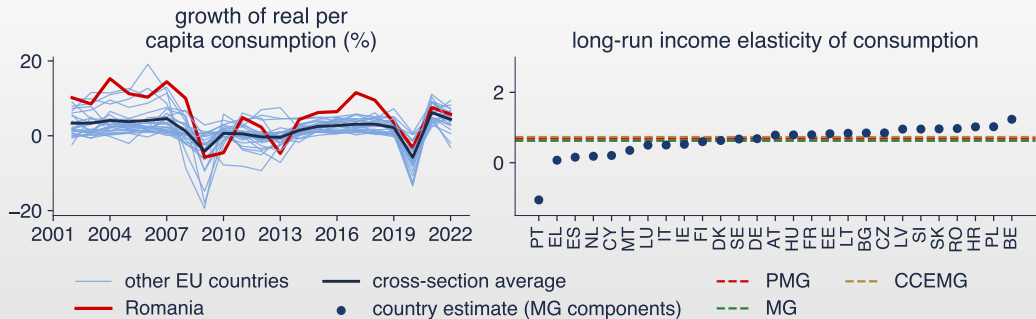
$$\Delta y_{it} = \phi_i(y_{i,t-1} - \theta' x_{i,t-1}) + \sum_j \lambda_{ij} \Delta y_{i,t-j} + \sum_j \delta'_{ij} \Delta x_{i,t-j} + \mu_i + \varepsilon_{it}$$

- ▶ θ : the long-run coefficients, common to all countries; ϕ_i : the country speed of adjustment, $\phi_i < 0$ required
- ▶ $\lambda_{ij}, \delta_{ij}, \mu_i, \sigma_i^2 = \text{Var} \varepsilon_{it}$: short run, intercept and error variance, all country-specific
- ▶ ML: maximise $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ over θ ; T_i : the sample length of country i (derivation: [Appendix](#))
- ▣ Hausman test of $\theta_{MG} = \theta_{PMG}$
 - ▶ $H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})' [\widehat{\text{Var}}(\hat{\theta}_{MG}) - \widehat{\text{Var}}(\hat{\theta}_{PMG})]^{-1} (\hat{\theta}_{MG} - \hat{\theta}_{PMG}) \rightarrow \chi^2(\dim \theta)$
 - ▶ PMG is efficient under homogeneity, MG consistent in both cases: a small H (large p-value) supports pooling

Case study: Pesaran, Shin and Smith (1999)

- Pesaran, Shin and Smith (1999)
 - ▶ consumption functions for OECD countries, annual data
 - ▶ ARDL in log real per capita consumption, log real per capita disposable income and inflation
 - ▶ they compare MG, PMG and dynamic fixed effects and use Hausman tests to decide whether the long-run income and inflation effects can be pooled
 - ▶ their argument: theory (permanent income) restricts the long run; adjustment speeds, habits and credit constraints differ across countries
- Our replication of the specification on EU-27, 2001–2022, ARDL(1,1,1) for every country, 672 observations
 - ▶ c_{it} on y_{it} and π_{it} as defined above; national currencies (country intercepts absorb the units)
- We start in 2000 to avoid the inflation episodes of the 1990s in Bulgaria and Romania, which would dominate a pooled inflation coefficient

Heterogeneity across the EU



- Left: consumption growth (Romania in red, cross-section average in dark blue); right: country long-run income elasticities from the individual ARDLs, with MG, PMG and CCEMG
- Country elasticities range from -1.06 to 1.23 ; Romania 0.97

Long-run estimates for EU-27

Estimator	θ_y	θ_π	ϕ
MG	0.63 (0.09)	0.82 (0.32)	-0.43 (0.032)
PMG	0.69 (0.04)	0.41 (0.18)	-0.31 (0.028)
dynamic FE	0.81 (0.03)	0.01 (0.23)	-0.29 (0.025)
group-mean DOLS	0.77 (0.05)	-0.24 (0.17)	—
CCEMG	0.73 (0.05)	0.10 (0.10)	—

- Standard errors in brackets (MG, DOLS, CCEMG: dispersion of country estimates; PMG: likelihood; FE: clustered by country)
- Hausman MG against PMG: $H = 2.41$, $\chi^2(2)$, $p = 0.300$; CD of the PMG residuals: 30.9

Interpreting the panel estimates

- The income elasticity lies between 0.63 (MG) and 0.81 (FE), clearly below one: consumption does not track income one for one over 2001–2022
- The Hausman test does not reject pooling, so PMG is the efficient choice for the income effect
- The inflation effect is not robust: its sign and size change with the estimator; do not report one of them alone
- PMG residuals remain strongly dependent ($CD = 30.9$): standard errors that ignore common shocks are too small; CCEMG is the safer benchmark

Panel VAR in one slide

- $y_{it} = \mu_i + \sum_{j=1}^p A_j y_{i,t-j} + \varepsilon_{it}$: common dynamics, country fixed effects [Holtz-Eakin, Newey and Rosen \(1988\)](#)
 - ▶ y_{it} : the vector of variables of country i ; A_j : lag matrices common to all countries; μ_i : country fixed effects; ε_{it} : innovations
- With small T the within estimator is biased by $O(1/T)$ (Nickell); estimate by GMM on forward-orthogonal deviations with lagged levels as instruments [Abrigo and Love \(2016\)](#)
- Identification of shocks as in [Chapter 3](#) (Cholesky, signs); impulse responses and FEVD are common to all countries
- Homogeneity of A_j is the strong assumption: with macro panels of moderate T , mean-group VARs or CCE-augmented VARs are the heterogeneous alternatives

Recap: Panel time series

- ▣ Test for cross-section dependence first; with strong dependence use CIPS and CCE, not LLC/IPS and pooled OLS
- ▣ Panel cointegration tests differ in their null and in the common-factor restriction; disagreement is informative
- ▣ MG for heterogeneity, PMG when theory pools the long run, Hausman to decide, CCE against common factors

An open question

- Is the long-run pass-through of money-market rates to bank lending rates complete in Romania, and did it change after the 2022 tightening?
 - ▶ formal: $H_0: \theta_{\text{lend}} = 1$ in a rank-2 VECM with ROBOR weakly exogenous; against a break in θ or in the markup in 2022
 - ▶ falsified by a robust rejection: across pre-registered lag lengths, deterministic cases and bootstrap inference
- Why it matters: the speed and completeness of pass-through decide how much the BNR must move its rate to move credit conditions
 - ▶ literature to start from: de Bondt (2005), Égert, Crespo-Cuaresma and Reininger (2007), Johansen (1995a), Cavaliere, Rahbek and Taylor (2012), Pesaran, Shin and Smith (2001)

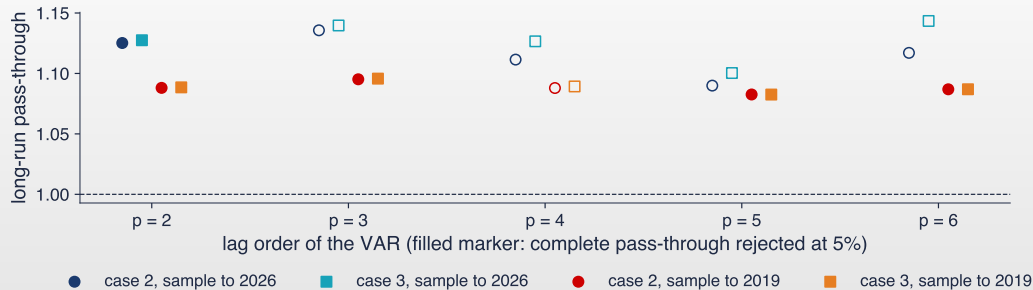
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed studies of interest-rate pass-through in Central and Eastern Europe that use cointegrated VARs or ARDL; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three reasons why lending-rate pass-through could exceed one, and one testable implication of each.
 - ▶ **code and replication:** ask for a Johansen function with restricted constant, then reproduce a known number first (the eigenvalues of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: list every specification choice that could flip the complete pass-through test.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The deterministic case of the critical values matches the estimated model (a frequent AI error: case-3 values for a case-2 model)
- The degrees of freedom of every LR test on β and α are counted by hand
- Lag length, case, sample and the treatment of 2008–2009 are fixed before the results; all variants are reported
- A claim of “complete pass-through” is checked against the confidence set of θ , not against one p -value

Mini-case: the estimate is robust, the verdict is not



- Long-run pass-through to the lending rate in the rank-2 VECM: lags $p = 2, \dots, 6$, cases 2 and 3, samples ending in 2019 and in 2026; filled markers: $\theta = 1$ rejected at 5%
- All 20 estimates lie between 1.08 and 1.14, yet complete pass-through is rejected in 10 of them: an AI summary that reports one p -value “with confidence” is wrong

Project idea

- **Interest-rate pass-through in Central and Eastern Europe:** replicate first, then extend
 - ▶ replicate: the Romanian rank-2 VECM of this lecture (eigenvalues 0.244, 0.177, 0.017) and the ARDL pass-through 1.15
 - ▶ extend: Romania, Hungary, Czechia and Poland; broken constants [Johansen, Mosconi and Nielsen \(2000\)](#) in 2008 and 2022; wild-bootstrap rank tests; PMG across countries with a Hausman test; NARDL for asymmetric pass-through
 - ▶ pre-register: variables, sample, case, lag rule, rank procedure, the restrictions to test and the multiple-testing correction
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Johansen = reduced-rank regression; the rank test is non-standard, depends on the case and over-rejects in small samples
- ▣ Given the rank, restrictions on β (economics) and α (weak exogeneity) are ordinary χ^2 tests
- ▣ Cointegration supplies long-run zeros for structural analysis: permanent and transitory shocks
- ▣ ARDL is a conditional VECM: valid under weak exogeneity and a single relation, with small-sample bounds
- ▣ Panels add power but bring heterogeneity and cross-section dependence: CD first, then CIPS, MG/PMG and CCE

Self-assessment

Questions

- Why does one eigen-decomposition give the ML estimator of β for every rank?
- Which deterministic case fits two interest rates without drift?
- How many degrees of freedom has the test of a known β with $n_1 = 3$, $r = 2$?
- What does weak exogeneity of ROBOR allow you to do?
- Why can IPS reject a unit root that CIPS does not reject?

Next:

- Bayesian VAR, factor models and nowcasting
- Shrinkage handles the parameter problem we met in small samples; factors generalise the cross-section averages of CCE

Appendix: from the likelihood to the eigenvalues

- ▣ For fixed β : $|\hat{\Omega}(\beta)| = |S_{00} - S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}|$; apply the partitioned determinant to $\begin{pmatrix} S_{00} & S_{01}\beta \\ \beta'S_{10} & \beta'S_{11}\beta \end{pmatrix}$ in both orders
- ▣ $|\hat{\Omega}(\beta)| |\beta'S_{11}\beta| = |S_{00}| |\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|$, which gives the ratio on the slide
- ▣ Write $\beta = S_{11}^{-1/2}\eta$: the ratio becomes $|\eta'(I - M)\eta|/|\eta'\eta|$ with $M = S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}$; it is minimised by the eigenvectors of M with the r largest eigenvalues
- ▣ The minimum is $\prod_{i \leq r} (1 - \lambda_i)$; nested ranks share eigenvectors, so the LR statistics are sums of $-T \ln(1 - \hat{\lambda}_i)$

Appendix: deriving the long-run impact matrix C

- ▣ Multiply the VECM by $\alpha'_{\perp}$: $\alpha'_{\perp} \Delta y_t = \alpha'_{\perp} \sum_i \Gamma_i \Delta y_{t-i} + \alpha'_{\perp} \varepsilon_t$; summing gives $\alpha'_{\perp} \Gamma y_t \approx \alpha'_{\perp} \sum_{s \leq t} \varepsilon_s$ up to stationary terms
- ▣ Decompose $y_t = \beta(\beta' \beta)^{-1} \beta' y_t + \beta_{\perp}(\beta'_{\perp} \beta_{\perp})^{-1} \beta'_{\perp} y_t$; the first part is stationary
- ▣ Hence $\alpha'_{\perp} \Gamma \beta_{\perp} (\beta'_{\perp} \beta_{\perp})^{-1} \beta'_{\perp} y_t \approx \alpha'_{\perp} \sum_s \varepsilon_s$; invert $\alpha'_{\perp} \Gamma \beta_{\perp}$ (the $I(1)$ condition) and premultiply by $\beta_{\perp}$
- ▣ Result: the non-stationary part of y_t is $\beta_{\perp} (\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp} \sum_s \varepsilon_s = C \sum_s \varepsilon_s$

Appendix: the PMG likelihood

$$\square \ell(\theta, \phi, \sigma^2) = - \sum_{i=1}^N \frac{T_i}{2} \ln(2\pi\sigma_i^2) - \sum_{i=1}^N \frac{1}{2\sigma_i^2} (\Delta y_i - \phi_i \xi_i(\theta))' H_i (\Delta y_i - \phi_i \xi_i(\theta))$$

□ Notation

- ▶ $\Delta y_i, y_{i,-1}$ ($T_i \times 1$), $X_{i,-1}$ ($T_i \times k$): the changes, the lagged levels of y and of x for country i
- ▶ $\xi_i(\theta) = y_{i,-1} - X_{i,-1}\theta$: the equilibrium error of country i
- ▶ $H_i = I - W_i(W_i'W_i)^{-1}W_i'$; W_i : the country's short-run regressors and intercept, which H_i projects out

- Given θ, ϕ_i and σ_i^2 are country OLS; the concentrated likelihood $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ is maximised over θ (back-substitution or Newton)
- $\hat{\theta}_{PMG}$ is consistent and asymptotically normal if $\phi_i < 0$ for all i and the long run is homogeneous; the $I(0)/I(1)$ status of x does not matter, as in ARDL

References (1/5)

- ▣ Abrigo, M. R. M., & Love, I. (2016). Estimation of panel vector autoregression in Stata. *The Stata Journal*, 16(3), 778–804. doi:10.1177/1536867X1601600314
- ▣ Bacon, R. W. (1991). Rockets and feathers: The asymmetric speed of adjustment of UK retail gasoline prices to cost changes. *Energy Economics*, 13(3), 211–218. doi:10.1016/0140-9883(91)90022-R
- ▣ Breitung, J., & Pesaran, M. H. (2008). Unit roots and cointegration in panels. In L. Mátyás & P. Sevestre (Eds.), *The Econometrics of Panel Data* (pp. 279–322). Springer. doi:10.1007/978-3-540-75892-1_9
- ▣ Cavaliere, G., Rahbek, A., & Taylor, A. M. R. (2010). Cointegration rank testing under conditional heteroskedasticity. *Econometric Theory*, 26(6), 1719–1760. doi:10.1017/S0266466609990776
- ▣ Cavaliere, G., Rahbek, A., & Taylor, A. M. R. (2012). Bootstrap determination of the co-integration rank in vector autoregressive models. *Econometrica*, 80(4), 1721–1740. doi:10.3982/ECTA9099
- ▣ Chudik, A., & Pesaran, M. H. (2015). Common correlated effects estimation of heterogeneous dynamic panel data models with weakly exogenous regressors. *Journal of Econometrics*, 188(2), 393–420. doi:10.1016/j.jeconom.2015.03.007
- ▣ de Bondt, G. J. (2005). Interest rate pass-through: Empirical results for the euro area. *German Economic Review*, 6(1), 37–78. doi:10.1111/j.1465-6485.2005.00121.x
- ▣ Égert, B., Crespo-Cuaresma, J., & Reininger, T. (2007). Interest rate pass-through in Central and Eastern Europe: Reborn from ashes merely to pass away? *Journal of Policy Modeling*, 29(2), 209–225. doi:10.1016/j.jpolmod.2007.01.005
- ▣ Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276. doi:10.2307/1913236
- ▣ Gonzalo, J., & Ng, S. (2001). A systematic framework for analyzing the dynamic effects of permanent and transitory shocks. *Journal of Economic Dynamics and Control*, 25(10), 1527–1546. doi:10.1016/S0165-1889(99)00062-7
- ▣ Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- ▣ Holtz-Eakin, D., Newey, W., & Rosen, H. S. (1988). Estimating vector autoregressions with panel data. *Econometrica*, 56(6), 1371–1395. doi:10.2307/1913103

References (2/5)

- Im, K. S., Pesaran, M. H., & Shin, Y. (2003). Testing for unit roots in heterogeneous panels. *Journal of Econometrics*, 115(1), 53–74. doi:10.1016/S0304-4076(03)00092-7
- Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control*, 12(2–3), 231–254. doi:10.1016/0165-1889(88)90041-3
- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580. doi:10.2307/2938278
- Johansen, S. (1992). Determination of cointegration rank in the presence of a linear trend. *Oxford Bulletin of Economics and Statistics*, 54(3), 383–397. doi:10.1111/j.1468-0084.1992.tb00008.x
- Johansen, S. (1995a). *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press. doi:10.1093/0198774508.001.0001
- Johansen, S. (1995b). Identifying restrictions of linear equations with applications to simultaneous equations and cointegration. *Journal of Econometrics*, 69(1), 111–132. doi:10.1016/0304-4076(94)01664-L
- Johansen, S. (1995c). A statistical analysis of cointegration for I(2) variables. *Econometric Theory*, 11(1), 25–59. doi:10.1017/S0266466600009026
- Johansen, S. (2002). A small sample correction for the test of cointegrating rank in the vector autoregressive model. *Econometrica*, 70(5), 1929–1961. doi:10.1111/1468-0262.00358
- Johansen, S., & Juselius, K. (1990). Maximum likelihood estimation and inference on cointegration, with applications to the demand for money. *Oxford Bulletin of Economics and Statistics*, 52(2), 169–210. doi:10.1111/j.1468-0084.1990.mp52002003.x
- Johansen, S., Mosconi, R., & Nielsen, B. (2000). Cointegration analysis in the presence of structural breaks in the deterministic trend. *The Econometrics Journal*, 3(2), 216–249. doi:10.1111/1368-423X.00047
- Juselius, K. (2006). *The Cointegrated VAR Model: Methodology and Applications*. Oxford University Press. doi:10.1093/oso/9780199285662.001.0001
- Kao, C. (1999). Spurious regression and residual-based tests for cointegration in panel data. *Journal of Econometrics*, 90(1), 1–44. doi:10.1016/S0304-4076(98)00023-2

References (3/5)

- ▣ Kapetanios, G., Pesaran, M. H., & Yamagata, T. (2011). Panels with non-stationary multifactor error structures. *Journal of Econometrics*, 160(2), 326–348. doi:10.1016/j.jeconom.2010.10.001
- ▣ Kilian, L., & Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press. doi:10.1017/9781108164818
- ▣ King, R. G., Plosser, C. I., Stock, J. H., & Watson, M. W. (1991). Stochastic trends and economic fluctuations. *American Economic Review*, 81(4), 819–840. [jstor.org](https://www.jstor.org)
- ▣ Kripfganz, S., & Schneider, D. C. (2020). Response surface regressions for critical value bounds and approximate p-values in equilibrium correction models. *Oxford Bulletin of Economics and Statistics*, 82(6), 1456–1481. doi:10.1111/obes.12377
- ▣ Levin, A., Lin, C.-F., & Chu, C.-S. J. (2002). Unit root tests in panel data: Asymptotic and finite-sample properties. *Journal of Econometrics*, 108(1), 1–24. doi:10.1016/S0304-4076(01)00098-7
- ▣ Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer. doi:10.1007/978-3-540-27752-1
- ▣ MacKinnon, J. G., Haug, A. A., & Michelis, L. (1999). Numerical distribution functions of likelihood ratio tests for cointegration. *Journal of Applied Econometrics*, 14(5), 563–577. doi:10.1002/(SICI)1099-1255(199909/10)14:5<563::AID-JAE530>3.0.CO;2-R
- ▣ Narayan, P. K. (2005). The saving and investment nexus for China: Evidence from cointegration tests. *Applied Economics*, 37(17), 1979–1990. doi:10.1080/00036840500278103
- ▣ Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6), 1417–1426. doi:10.2307/1911408
- ▣ Osterwald-Lenum, M. (1992). A note with quantiles of the asymptotic distribution of the maximum likelihood cointegration rank test statistics. *Oxford Bulletin of Economics and Statistics*, 54(3), 461–472. doi:10.1111/j.1468-0084.1992.tb00013.x
- ▣ Pantula, S. G. (1989). Testing for unit roots in time series data. *Econometric Theory*, 5(2), 256–271. doi:10.1017/S0266466600012421
- ▣ Pedroni, P. (1999). Critical values for cointegration tests in heterogeneous panels with multiple regressors. *Oxford Bulletin of Economics and Statistics*, 61(S1), 653–670. doi:10.1111/1468-0084.0610s1653

References (4/5)

- ▣ Pedroni, P. (2001). Purchasing power parity tests in cointegrated panels. *The Review of Economics and Statistics*, 83(4), 727–731. doi:10.1162/003465301753237803
- ▣ Pedroni, P. (2004). Panel cointegration: Asymptotic and finite sample properties of pooled time series tests with an application to the PPP hypothesis. *Econometric Theory*, 20(3), 597–625. doi:10.1017/S0266466604203073
- ▣ Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4), 967–1012. doi:10.1111/j.1468-0262.2006.00692.x
- ▣ Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265–312. doi:10.1002/jae.951
- ▣ Pesaran, M. H. (2015a). Testing weak cross-sectional dependence in large panels. *Econometric Reviews*, 34(6–10), 1089–1117. doi:10.1080/07474938.2014.956623
- ▣ Pesaran, M. H. (2015b). *Time Series and Panel Data Econometrics*. Oxford University Press. doi:10.1093/acprof:oso/9780198736912.001.0001
- ▣ Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*, 60(1), 13–50. doi:10.1007/s00181-020-01875-7
- ▣ Pesaran, M. H., & Shin, Y. (1998). An autoregressive distributed-lag modelling approach to cointegration analysis. In S. Strøm (Ed.), *Econometrics and Economic Theory in the 20th Century: The Ragnar Frisch Centennial Symposium* (pp. 371–413). Cambridge University Press. doi:10.1017/CCOL0521633230.011
- ▣ Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of Applied Econometrics*, 16(3), 289–326. doi:10.1002/jae.616
- ▣ Pesaran, M. H., Shin, Y., & Smith, R. P. (1999). Pooled mean group estimation of dynamic heterogeneous panels. *Journal of the American Statistical Association*, 94(446), 621–634. doi:10.1080/01621459.1999.10474156
- ▣ Pesaran, M. H., & Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, 68(1), 79–113. doi:10.1016/0304-4076(94)01644-F
- ▣ Phillips, P. C. B., & Hansen, B. E. (1990). Statistical inference in instrumental variables regression with $I(1)$ processes. *The Review of Economic Studies*, 57(1), 99–125. doi:10.2307/2297545

References (5/5)

- ▣ Phillips, P. C. B., & Moon, H. R. (1999). Linear regression limit theory for nonstationary panel data. *Econometrica*, 67(5), 1057–1111. doi:10.1111/1468-0262.00070
- ▣ Reinsel, G. C., & Ahn, S. K. (1992). Vector autoregressive models with unit roots and reduced rank structure: Estimation, likelihood ratio test, and forecasting. *Journal of Time Series Analysis*, 13(4), 353–375. doi:10.1111/j.1467-9892.1992.tb00113.x
- ▣ Saikkonen, P. (1991). Asymptotically efficient estimation of cointegration regressions. *Econometric Theory*, 7(1), 1–21. doi:10.1017/S0266466600004217
- ▣ Shin, Y., Yu, B., & Greenwood-Nimmo, M. (2014). Modelling asymmetric cointegration and dynamic multipliers in a nonlinear ARDL framework. In R. C. Sickles & W. C. Horrace (Eds.), *Festschrift in Honor of Peter Schmidt* (pp. 281–314). Springer. doi:10.1007/978-1-4899-8008-3_9
- ▣ Stock, J. H., & Watson, M. W. (1993). A simple estimator of cointegrating vectors in higher order integrated systems. *Econometrica*, 61(4), 783–820. doi:10.2307/2951763
- ▣ Swensen, A. R. (2006). Bootstrap algorithms for testing and determining the cointegration rank in VAR models. *Econometrica*, 74(6), 1699–1714. doi:10.1111/j.1468-0262.2006.00723.x
- ▣ Toda, H. Y., & Yamamoto, T. (1995). Statistical inference in vector autoregressions with possibly integrated processes. *Journal of Econometrics*, 66(1–2), 225–250. doi:10.1016/0304-4076(94)01616-8
- ▣ Westerlund, J. (2007). Testing for error correction in panel data. *Oxford Bulletin of Economics and Statistics*, 69(6), 709–748. doi:10.1111/j.1468-0084.2007.00477.x